

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years building event-driven systems and real-time data pipelines in Node.js and TypeScript. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, implementing event-driven architecture using Pub/Sub for real-time data synchronization. Experienced in designing microservices with Apache Kafka, AWS cloud services (S3, EC2), and clean architecture principles. Proven track record of cross-functional ownership from architecture design to production deployment, including leading a team of 4 engineers delivering enterprise systems for legal tech NGOs.

CORE COMPETENCIES

TypeScript / Node.js	Event-Driven Architecture	AWS Cloud Services	Apache Kafka	Microservices	Clean Architecture
CI/CD Pipelines	Real-Time Data Systems				

WORK EXPERIENCE

MTailor September 2022 – Present

Software Engineer (US Remote)

- **Led end-to-end migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO
- Implemented **event-driven architecture** using Node.js and Google Pub/Sub for real-time two-way synchronization between MongoDB and Firestore
- Built a 3D visualization feature for customers using video overlay and the ffmpeg library, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via Python automation scripts, demonstrating multi-cloud infrastructure expertise
- Implemented Express Shipping feature, generating an additional \$40 revenue per order
- Reduced infrastructure costs by \$5,000/month by migrating all AWS services to GCP

MTailor March 2022 – July 2022

Contractor (US Remote)

- Executed complex backend migration from Parse/MongoDB to Firebase Firestore and GCP, hired via Upwork for specialized expertise
- Built migration scripts in Node.js to read from MongoDB and write to Firestore with complex data transformation logic
- Wrote technical documentation for engineers on Firebase SDKs (Node.js and Python) and trained Ops team on new infrastructure

Mind2matter.co January 2022 – July 2022

Software Engineer (US Remote)

- Built backends for DeFi applications using Web3 and Node.js for a US-based agency, reporting directly to CTO
- Delivered React UIs from Figma designs under tight timelines, demonstrating full-stack capabilities

Dr Wealth August 2021 – January 2022

Software Engineer (Singapore Remote)

- Optimized PWA delivering real-time stock market prices from a Firebase backend serving high-throughput data
- Built ad-hoc jobs to query over 2 million Firestore records, keeping customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku

CodeBits January 2020 – July 2021

Software Engineer (Uganda Remote)

- **Led team of 4 developers** building legal tech systems for NGOs in Uganda, demonstrating mentorship and technical leadership
- Architected **microservices backend with Apache Kafka** for FIDA Uganda's case management application, implementing event-driven patterns with Docker and Kubernetes
- Built case management system for FIDA Uganda and paralegal database for LASPNET Uganda
- Developed cross-platform mobile app using React Native and Expo (iOS & Android) with Firebase push notifications
- Built USSD service for legal aid providers communicating with Node.js backend via gRPC, enabling rural accessibility

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)
Kampala, Uganda

2018 – 2022

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC
Databases: MongoDB, Firestore, DynamoDB, SQL **Cloud:** AWS (S3, EC2, SQS), GCP (Pub/Sub, Cloud Functions), Firebase
Event Streaming: Apache Kafka, Google Pub/Sub, NATS Streaming **DevOps:** Docker, Kubernetes, CI/CD, Infrastructure as Code
Frontend: React, React Native, HTML, CSS, Tailwind